

DUET: Endogenous Failure Detection and Recovery via Dual-Action Manifold Alignment

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Abstract—Vision-language-action (VLA) models map vision and language to robot actions. With limited post-training demonstrations, however, they often cannot detect or recover from execution failures that move the robot out of distribution (OOD), such as mis-grasps, dropped objects, unexpected occlusions, and object or robot displacement. Existing solutions rely on external failure detectors and costly intervention or recovery data, leaving detection loosely coupled with control. We introduce DUET (Dual-manifold Unified Execution and Transfer), a dual-action-head module that predicts relative and absolute actions. DUET uses the endogenous disagreement between the two heads as its primary failure signal, complemented by state-distribution distance. The absolute head excludes the current robot state and predicts RVQ-anchored global waypoints from vision and language, providing stable recovery targets when relative control becomes unreliable. We evaluate DUET on four LIBERO suites under five families of execution failures, and transfer the identical module to a π_0 backbone on a real Franka arm. DUET preserves nominal success (96.9% vs. 93.8% for the relative-only baseline) while raising success under action-execution failures from 27.9% to 64.6% (+36.7 pp). The same disagreement score leads all success-only detectors on AUROC (0.845). On hardware, DUET matches π_0 under nominal execution (90.0%) and raises robot-state OOD success from 10.0% to 77.5%.

I. INTRODUCTION

Vision-language-action (VLA) policies adapt pretrained vision-language models to map visual observations and language instructions directly to robot actions [1], [2], [3]. Despite their semantic generalization, policies post-trained on limited robot demonstrations remain vulnerable to failures during closed-loop execution. We focus on failures that drive the visual observation, robot–object configuration, or induced trajectory out of distribution (OOD) relative to successful demonstrations. They include policy-induced errors, such as an off-center grasp or an accidentally dropped object, as well as external events, such as transient occlusion or unexpected displacement of an object or the robot arm. After such events, a VLA may continue to produce locally plausible actions even though its trajectory is no longer supported by the demonstrations. Detecting these failures and actively recovering from them can therefore substantially improve task success and deployment reliability.

Existing approaches commonly introduce an external monitor, critic, verifier, or predictive model for failure detection [4], and rely on large amounts of human takeover data collected during failed executions to learn recovery [5], [6], [7]. This process is costly because operators must continuously monitor rollouts and provide corrective actions across diverse

visual, object, and robot-state failures. Moreover, detection and recovery are learned separately, so a failure alarm is not necessarily aligned with the action needed to recover. We instead seek a single policy that uses endogenous action information to detect and recover from OOD failures without additional failure or intervention data.

DUET takes disagreement between different action spaces as an endogenous signal for failure detection, as illustrated in Fig. 1. In our preliminary experiments, policies trained with relative actions produced smoother and more stable motions during nominal execution, but struggled to recover once a grasp or motion deviated because state-relative commands provide no explicit return target. In contrast, policies trained with absolute actions could redirect the robot toward trajectories supported by the demonstration distribution and retry after a failure. This complementary behavior motivates DUET to learn the same demonstrated behavior with a relative action head and an absolute action head. The relative head serves as the nominal controller, while the absolute head provides global recovery anchors. The two action spaces agree under in-domain execution but necessarily diverge when a failure or OOD event drives the robot away from the demonstrated trajectory, thereby coupling failure detection and recovery through the same signal.

At inference time, DUET converts the predicted absolute waypoint into a correction from the live pose and compares it with the relative action. The two predictions remain consistent along successful trajectories but diverge when a failure drives execution OOD. Their disagreement, together with distance from the training state distribution, determines a soft fusion weight: DUET follows the smooth relative controller during nominal execution, shifts toward the absolute recovery target after failure, and returns to relative control after re-entering the demonstrated support. The same cross-space signal therefore both detects failure and determines the recovery direction, except when the observation itself is invalid and a cached waypoint must be held.

Our contributions are threefold:

- We propose DUET, a dual-action-head VLA that learns relative and absolute action spaces from successful demonstrations and uses their endogenous disagreement to detect OOD-induced failures without an external detector.
- We design a stateless, RVQ-anchored absolute action head and an adaptive closed-loop controller that converts cross-space disagreement into stable recovery while preserving precise relative control under nominal conditions.

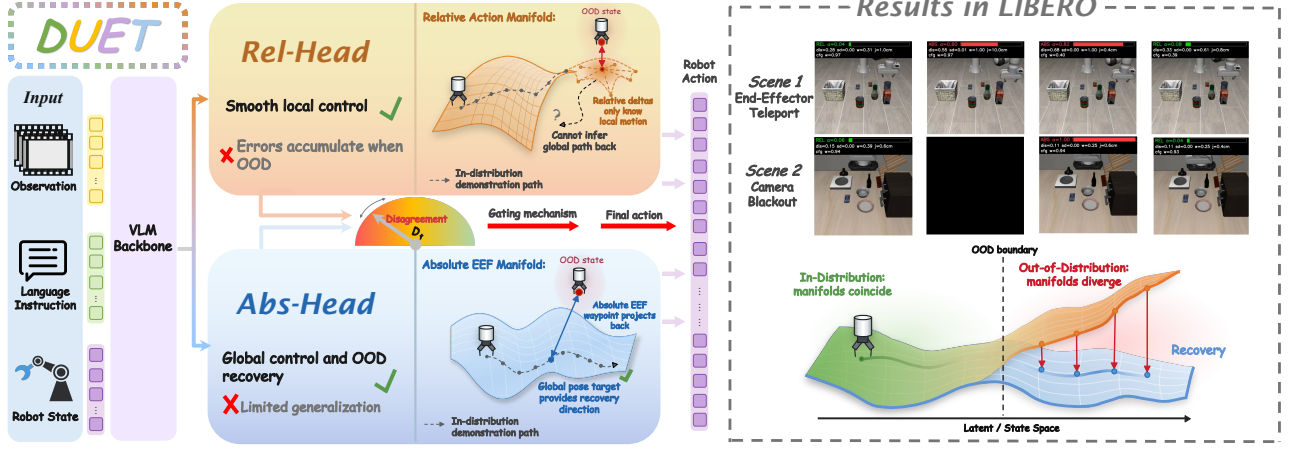


Fig. 1. Dual-manifold failure detection and recovery. The relative and absolute end-effector manifolds agree during nominal execution but diverge when a failure drives the system OOD, exposing an endogenous disagreement score D_t . Relative actions provide precise local motion, whereas absolute waypoints project execution toward a stable demonstrated-manifold anchor.

- We validate DUET on four LIBERO suites under five failure families—action-execution drift, robot displacement, grasp loss, transient occlusion, and external contact—and transfer the same module to a π_0 backbone on a real Franka arm.

II. RELATED WORK

A. Failure Detection for Robot Policies

Runtime failure detection estimates whether a policy remains within the support of successful behavior. Diff-Dagger uses diffusion uncertainty to request expert intervention [8], while Sentinel monitors action inconsistency and task progress [9]. FAIL-Detect formulates failure detection as sequential OOD testing using only successful executions [4]; FIPER similarly combines representation novelty with action uncertainty without failure data [10].

VLA-specific methods use internal features or external foundation models. SAFE learns a detector from successful and failed rollouts [11], whereas AHA trains a VLM to detect and diagnose failures [12]. Such methods require additional supervision or a separate detector whose output is not a recovery action. DUET instead detects failures through disagreement between two action spaces learned within the policy, using OOD manifold inconsistency as both an alarm and a recovery signal.

B. Recovery from Execution Failures

Recovery methods commonly expand the training distribution with expert corrections. DAgger queries experts on policy-induced states [5]; intervention-based methods collect human takeovers [13], [6]; and IntervGen transforms limited interventions into corrective trajectories [7]. These approaches reduce compounding errors but still require additional recovery actions.

Foundation-model systems instead use semantic supervisors. RACER learns from language-annotated recovery trajectories [14], while AHA and VLM controllers diagnose

failures and guide downstream correction [12], [15]. Their detection and recovery components remain separately trained or prompted. DUET couples both capabilities without extra recovery data: absolute states from successful demonstrations provide recovery anchors, while cross-space disagreement determines when and how strongly to move toward them.

III. METHOD

A. Problem Formulation

At time t , the robot observes multi-view images o_t , a language instruction l , and its current proprioceptive state s_t . Trained only on successful demonstrations, DUET predicts the demonstrated behavior in two action representations over a horizon K :

$$\hat{\Delta}A_t = f_{\text{rel}}(o_t, l, s_t) \in \mathbb{R}^{K \times 7}, \quad (1)$$

$$\hat{W}_t = f_{\text{abs}}(o_t, l) \in \mathbb{R}^{K \times 7}. \quad (2)$$

Here, ΔA_t contains relative operational-space-control commands and W_t contains future absolute end-effector waypoints $[x, y, z, r_x, r_y, r_z, g]$. Each command decomposes as $a_t = [a_{p,t}, a_{g,t}]$, with pose $a_{p,t} \in \mathbb{R}^6$ and gripper $a_{g,t} \in \mathbb{R}$. A successful trajectory supplies both labels: recorded delta actions supervise f_{rel} , and future end-effector states supervise f_{abs} . Geometry and fusion operate on a_p ; the absolute head predicts gripper state in training, but the deployed a_g always comes from the relative head.

B. Dual Action Manifolds

For an instruction l , let $\mathcal{M}_l \subset SE(3)$ denote the end-effector manifold traced by successful demonstrations, and let $T\mathcal{M}_l$ denote its tangent bundle. Under nominal execution, the relative head predicts a local motion vector

$$\hat{\delta}a_{p,t} = f_{\text{rel}}(o_t, l, s_t)_p \in T_{p_t}\mathcal{M}_l, \quad (3)$$

where p_t is the current end-effector pose. This tangent representation is locally smooth, but specifies only how to move from the current chart. Once a failure leaves the

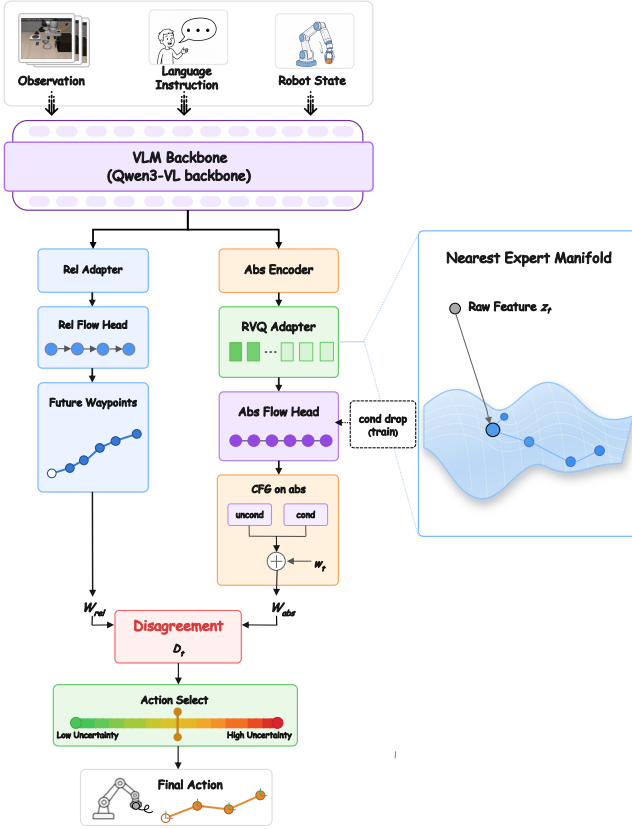


Fig. 2. DUET in four stages: a shared vision-language encoder, relative and absolute action heads, cross-space failure detection, and closed-loop recovery. The relative head provides state-aware local control; the state-free absolute head predicts RVQ-anchored global waypoints. Their disagreement gates the switch between nominal execution and recovery.

demonstrated support, the extrapolated field has no explicit return destination.

The absolute head instead predicts a future coordinate

$$\hat{w}_{p,t} = f_{\text{abs}}(o_t, l)_p \in \mathcal{M}_l, \quad (4)$$

which serves as a vision-language-conditioned global anchor. To compare the two representations, a state-dependent transport map

$$c_t = \Psi_{p_t}(\hat{w}_{p,t}) \in T_{p_t}SE(3), \quad (5)$$

converts the absolute waypoint into a local correction at the live pose. Along an in-domain trajectory, c_t and $\hat{\delta}a_{p,t}$ describe the same motion in two coordinates and therefore align. When execution is driven OOD, the relative head extrapolates a local field while the absolute head remains anchored to a demonstrated destination, exposing a cross-space inconsistency.

This geometry couples detection and recovery without a formal OOD guarantee: the inconsistency norm is a scalar failure signal, and the transported vector c_t is already the recovery direction. The same pair of predictions therefore decides both when and where the controller moves.

C. Dual-Head Architecture

Figure 2 shows the shared VLM and two flow-matching action heads [16], [17]. The relative head attends to the full

hidden sequence and receives s_t through a numerical state encoder, making it the state-aware nominal controller. The absolute head receives no explicit proprioception. When state tokens are appended to the causal VLM prompt, they are placed last and excluded from absolute-branch pooling, while the relative head can still attend to them. This prevents the absolute predictor from collapsing into an integration of the current state and preserves it as a global reference.

The absolute branch pools its state-free hidden sequence H^0 and applies residual vector quantization (RVQ) [18], [19]:

$$Z_q = \text{RVQ}(\text{Pool}(H^0)). \quad (6)$$

RVQ maps continuous conditions to a discrete atlas learned from successful demonstrations, anchoring absolute predictions to recurrent visual-language modes rather than an unconstrained feature. The residual norm is auxiliary telemetry and is not part of the primary failure score. The absolute flow head also uses classifier-free guidance (CFG) [20], softening the conditional prediction toward the demonstrated waypoint marginal during recovery.

D. Disagreement-Preserving Learning

Both heads are supervised by the paired labels in every successful trajectory, using flow-matching objectives with an RVQ commitment loss:

$$\mathcal{L} = \lambda_{\text{rel}}\mathcal{L}_{\text{rel}} + \lambda_{\text{abs}}\mathcal{L}_{\text{abs}} + \lambda_{\text{vq}}\mathcal{L}_{\text{vq}}, \quad (7)$$

where absolute labels are normalized future states and relative labels are recorded delta commands. We impose no cross-head consistency loss or distillation: forcing agreement would suppress the inconsistency needed for detection.

The branches are instead specialized by asymmetric visual augmentation. The absolute head and RVQ adapter train on both clean and perturbed observations so that the global anchor remains stable under degraded input. The relative head is optimized only on the clean subset $\mathcal{B}_{\text{clean}}$:

$$\mathcal{L}_{\text{rel}} = \frac{1}{|\mathcal{B}_{\text{clean}}|} \sum_{i \in \mathcal{B}_{\text{clean}}} \ell_{\text{FM}}(\hat{\Delta}a^{(i)}, \Delta a^{(i)}). \quad (8)$$

This gradient truncation keeps the relative head sensitive while training the absolute branch as a robust reference, making disagreement more discriminative under visual OOD. For CFG, the absolute head replaces its quantized condition with learned null tokens on a subset of samples and learns both conditional and marginal waypoint flows.

E. Joint Failure Detection and Recovery

At deployment, the policy returns both unnormalized action chunks. Let the live pose be $p_t = (x_t, R_t)$ and the pose component of the current absolute waypoint be $\hat{w}_{p,t} = (\hat{x}_t, \hat{R}_t)$. The transport map in Eq. (5) is

$$c_t = \text{clip}\left(\left[\frac{(\hat{x}_t - x_t)/\sigma_p}{\text{Log}(\hat{R}_t R_t^{-1})/\sigma_R}\right], -1, 1\right), \quad (9)$$

where $\text{Log} : SO(3) \rightarrow \mathfrak{so}(3)$ is the geodesic rotation correction and σ_p, σ_R convert pose error into action units. Cross-space disagreement is the primary failure signal,

$$D_t = \frac{\|c_t - \text{clip}(\hat{\delta}a_{p,t}, -1, 1)\|_2}{\sqrt{6}}. \quad (10)$$

We add a coarse state-distance term that measures normalized violation of the training-state bounds:

$$S_t = \text{mean} \left[\frac{\max(s_{\min} - p_t, 0)}{s_{\max} - s_{\min}} + \frac{\max(p_t - s_{\max}, 0)}{s_{\max} - s_{\min}} \right]. \quad (11)$$

The failure score and recovery weight are

$$q_t = D_t + \lambda S_t, \quad (12)$$

$$\alpha_t = \sigma(k(q_t - \tau)). \quad (13)$$

DUET interpolates between local execution and global recovery while keeping relative gripper control:

$$a_t = [a_{p,t}, a_{g,t}], \quad (14)$$

$$a_{p,t} = (1 - \alpha_t)\hat{\delta}a_{p,t} + \alpha_t c_t, \quad (15)$$

$$a_{g,t} = \hat{\delta}a_{g,t}, \quad (16)$$

where $[\hat{\delta}a_{p,t}, \hat{\delta}a_{g,t}]$ is the relative-head command. Relative control therefore dominates on the demonstrated support; a rising failure score shifts a_p toward the absolute recovery direction, and control returns after the two representations realign. When a new chunk is requested, the previous fusion weight also softens the absolute-head CFG scale,

$$w_t = w_{\max} - (w_{\max} - w_{\min})\alpha_{t-1}. \quad (17)$$

Thus D_t determines *when* to recover and c_t determines *how*. One exception is treated separately: a fully invalid observation corrupts both heads, so D_t is uninformative. A validity guard then freezes the last reliable waypoint and holds $\alpha_t=1$ until vision returns.

IV. EXPERIMENTS

We organize the simulation experiments around three questions:

- (Q1) Endogenous OOD detection:** Can DUET detect deviations using successful demonstrations only?
- (Q2) Nominal execution:** Does dual-action control preserve normal execution?
- (Q3) Failure-aware retry:** Can a detected, recoverable deviation be converted into task success?

We report detection, nominal success, and closed-loop retry separately, followed by mechanism studies and ablations.

A. Evaluation Protocol

We evaluate the four LIBERO suites [21] with 10 tasks and 50 initial states per task (500 episodes per suite), using suite-specific 10k-step checkpoints. Successful demonstrations provide future end-effector states for the absolute head and delta OSC commands for the relative head.

StarVLA [3] is the external relative-only control. DUET-Rel ($\alpha_t=0$) and DUET share a checkpoint, so their difference

TABLE I
EXECUTION-FAILURE FAMILIES AND THE QUANTITY DRIVEN OUT OF DISTRIBUTION.

Family	Protocol	What becomes OOD
Action execution	Action perturbation (step 5)	Command-to-motion map
Robot displacement	EEF teleport (step 39)	Robot-arm pose
Object / grasp	Vertical grasp drop (step 67)	Robot-object configuration
Visual occlusion	Camera blackout (step 23)	Current visual observation
Contact / scene	External force	Robot-world interaction

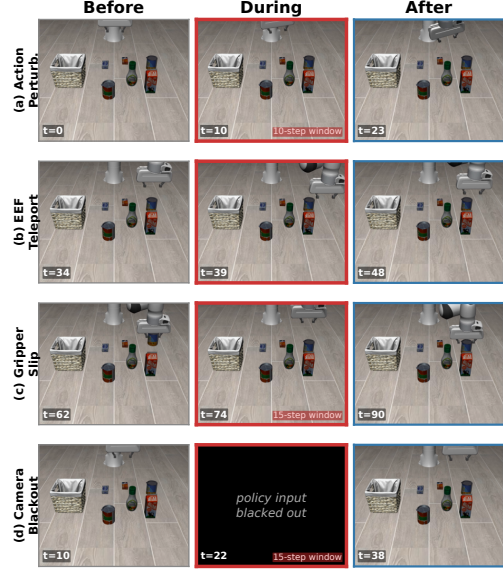


Fig. 3. Object-suite keyframes before, during, and after (a) action perturbation, (b) EEF teleport, (c) grasp drop, and (d) camera blackout.

isolates adaptive retry; StarVLA→DUET-Rel measures co-training. Absolute-only ($\alpha_t=1$) and fixed fusion ($\alpha_t=0.5$) are inference ablations.

DUET uses Qwen-VL features (size 2048), 16-/8-layer relative/absolute DiT-B heads, and 16 RVQ tokens from three codebooks (size 512, dimension 256). We set $p_{\text{aug}}=0.3$, $p_{\text{drop}}=0.15$, $\lambda=1$, $\tau=0.5$, and $k=8$.

Perturbations use fixed steps, deterministic seeds, and matched (task, initial state) keys; success differences use McNemar p -values. We retain uninjected episodes and report coverage for semantic preconditions. Table I defines the families; external force is reported separately and families are not pooled.

B. (Q1) Endogenous OOD Detection

For **Q1**, Table II compares D_t with matched success-only detectors [4], [9]. Unlike a separate alarm, D_t also activates correction c_t .

With delay close to prior methods (1.41 s versus 1.01–1.31 s), DUET achieves the highest AUROC (0.845) and TPR (0.900). Its balanced accuracy (0.582) remains below RND (0.602), so the gain is not uniform across thresholded metrics. Thus **Q1** is supported: disagreement improves failure ranking and detection without failure labels at modest delay.

TABLE II

FAILURE DETECTION UNDER MATCHED PROTOCOLS. ALL METRICS ARE HIGHER-IS-BETTER EXCEPT DELAY; BOLD MARKS THE BEST VALUE PER COLUMN.

Method	TPR $\uparrow$	TNR $\uparrow$	Bal.-Acc. $\uparrow$	AUROC $\uparrow$	Delay (steps) $\downarrow$	Delay (s) $\downarrow$	Loc.@16 $\uparrow$
DUET (D_t)	0.900	0.265	0.582	0.845	28.3	1.41	0.523
logpZO [4]	0.889	0.243	0.566	0.817	22.7	1.14	0.569
RND [22], [4]	0.842	0.361	0.602	0.549	26.1	1.31	0.330
PCA-kmeans [4]	0.884	0.283	0.583	0.625	20.3	1.01	0.503
STAC [9]	0.839	0.311	0.575	0.494	25.8	1.29	0.260

TABLE III

NOMINAL LIBERO SUCCESS (%). EXTERNAL ROWS ARE CONTEXT; STARVLA-GR00T USES JOINT 30K TRAINING AND DUET USES SUITE-SPECIFIC 10K CHECKPOINTS.

Method	Spatial	Object	Goal	LIBERO-10	Avg.
$\pi_{0.5}$ (reported)	98.8	98.2	98.0	92.4	96.9
StarVLA-GR00T (reported)	97.8	98.8	97.4	92.0	96.5
DUET	98.8	99.4	97.6	91.8	96.9
DUET-Rel ($\alpha_t=0$)	98.0	99.2	92.0	86.0	93.8

C. (Q2) Nominal Execution

For **Q2**, adaptive fusion raises the same checkpoint from 93.8% to 96.9% (Table III), with +5.6 pp on Goal and +5.8 pp on LIBERO-10. External rows provide context only. DUET therefore preserves nominal execution while correcting occasional clean-rollout drift.

D. (Q3) Failure-Aware Retry

For **Q3**, Table IV shows that DUET raises shared-checkpoint success from 27.9% to 64.6% under action perturbation, 34.5% to 59.4% after teleport, 89.2% to 95.2% after grasp drop, and 73.5% to 90.8% during blackout. Every aggregate also exceeds StarVLA.

Action execution. The intact scene offers a return target that relative control lacks. DUET gains 36.7 pp over 4,000 pairs; under severe shifts it retains 43.2–60.6%, versus 5.4–12.8% for Rel.

Robot displacement. Teleport shifts the live pose off-trajectory; DUET gains 24.9 pp overall and 39.8/47.6 pp on Object. Spatial is not significant ($p=0.21/0.067$), partly because episodes can finish before step 39.

Object and grasp. A 5 cm drop changes the robot–object relation but not the arm trajectory. The +6.0 pp gain ($p=2.4 \times 10^{-5}$; coverage $\geq 98.8\%$) over a strong 89.2% Rel baseline makes this a boundary case.

Visual occlusion. A blank image corrupts both heads, so the validity guard, not disagreement, drives recovery. DUET gains 17.3 pp; Object medium is the only regression (-1.8 pp, $p>0.05$), while LIBERO-10 severe gains 60.6 pp.

Contact and scene. External force is separate: DUET reaches 95.95% versus 92.22% over 4,000 episodes, with no suite regression.

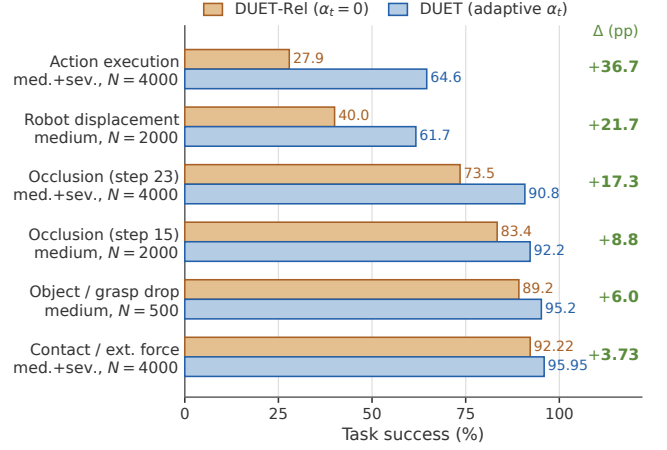


Fig. 4. Paired DUET-Rel→DUET success gain by family.

a) Mechanism evidence. Figure 5 shows switch-and-return under arm-state OOD: disagreement raises α_t , activates the waypoint correction, then falls after realignment. The gate stays weak for world-state change and is overridden during complete visual dropout.

On Object teleport (Fig. 6; $n=484-498$), DUET closes 4.69 cm by +32 steps and re-enters in 97.9% (median 16 steps), versus 0.71 cm, 62.5%, and 58 steps for Rel. Absolute-only reaches 99.6% re-entry but zero task success, so re-entry is mechanism evidence rather than the endpoint.

Together, final success and switch-and-return answer **Q3**: DUET helps when deviation is detectable and re-entry remains feasible; grasp changes and invalid vision bound disagreement alone.

E. Ablations

Table V tests fusion. Absolute-only loses precise execution; fixed fusion damages clean performance and under-corrects disturbances. Only adaptive fusion succeeds in both regimes.

Table VI isolates other choices. Removing RVQ disables the absolute branch; symmetric augmentation lowers clean and OOD success; adaptive CFG matters off-distribution.

a) Fusion on clean trajectories. Across 2,000 clean episodes, mean $\alpha_t=0.139$, its 95th percentile is 0.339, and only 1.7% of steps exceed 0.5, showing that relative control remains dominant.

TABLE IV

CLOSED-LOOP SUCCESS (%; $N=500$ PER CELL). OVERALL COMBINES MEDIUM+SEVERE ACROSS SUITES. REL→DUET HAS MCNEMAR $p<10^{-3}$ EXCEPT TELEPORT/SPATIAL ($p=0.21/0.067$) AND BLACKOUT/OBJECT MEDIUM.

	Medium				Severe			
Suite	StarVLA	Rel.	DUET	Δ	StarVLA	Rel.	DUET	Δ
<i>(i) Action perturbation (step 5)</i>								
Spatial	60.0	54.0	74.0	+20.0	14.0	12.8	43.2	+30.4
Object	41.6	48.8	84.4	+35.6	7.8	7.8	60.6	+52.8
Goal	52.2	47.2	74.2	+27.0	18.8	10.6	52.6	+42.0
LIBERO-10	36.2	36.6	78.6	+42.0	6.2	5.4	49.0	+43.6
Overall	29.6	27.9	64.6	+36.7		<i>med+sev combined</i>		
<i>(ii) EEf teleport (step 39)</i>								
Spatial	55.4	54.6	57.6	+3.0	46.8	48.2	52.6	+4.4
Object	43.4	36.2	76.0	+39.8	21.2	20.8	68.4	+47.6
Goal	41.6	33.4	52.4	+19.0	30.4	28.2	50.6	+22.4
LIBERO-10	40.4	35.8	60.8	+25.0	21.4	18.4	56.8	+38.4
Overall	37.6	34.5	59.4	+24.9		<i>med+sev combined</i>		
<i>(iii) Vertical grasp drop (Object only, medium, step 67)</i>								
Object	90.6	89.2	95.2	+6.0		—		
<i>(iv) Camera blackout (step 23)</i>								
Spatial	81.0	88.0	94.4	+6.4	42.4	63.2	95.2	+32.0
Object	91.8	98.6	96.8	−1.8	64.2	95.0	97.2	+2.2
Goal	65.4	86.0	86.4	+0.4	35.4	65.8	85.8	+20.0
LIBERO-10	60.0	65.6	84.2	+18.6	24.4	25.6	86.2	+60.6
Overall	58.1	73.5	90.8	+17.3		<i>med+sev combined</i>		

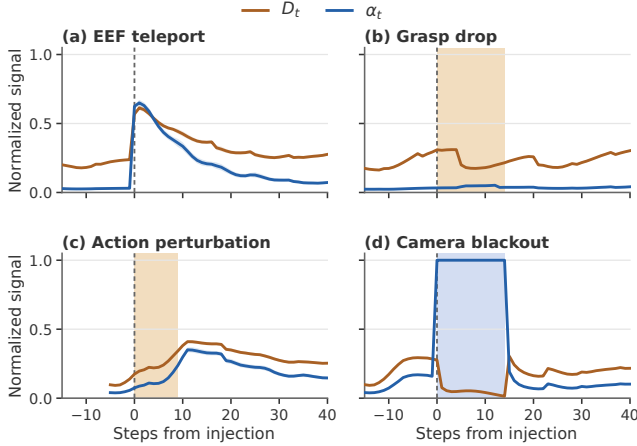


Fig. 5. D_t and α_t on Object, aligned to injection (mean, pointwise 95% CI): arm-state OOD (a,c), world-state change (b), and validity-guard override (d).

b) Sensitivity to τ and k .: Figure 7 shows a broad Object plateau above 90% clean and 75% OOD success. The shared $\tau=0.5$, $k=8$ lies inside it.

F. Real-Robot Evaluation

DUET attaches only to the shared vision-language representation, so the same heads, fusion rule, and validity guard transfer from StarVLA to π_0 (OpenPi) [17] without architectural change.

a) Platform and protocol.: We deploy on a 7-DoF Franka Emika Panda with a parallel-jaw gripper, a wrist-mounted Intel RealSense D405, and a third-person Orbbec FemtoBot camera, controlled at 20 Hz via delta joint-position

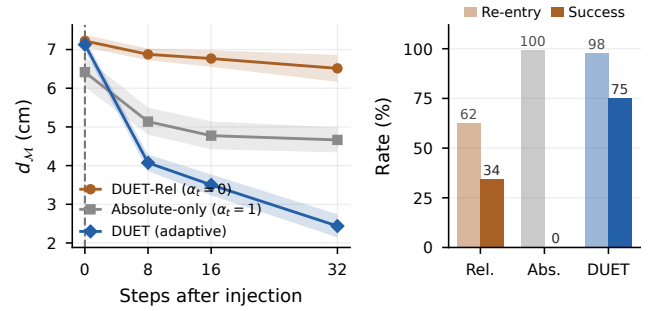


Fig. 6. Object EEf teleport (medium, step 39; $n=484-498$). Left: distance to the clean-trajectory reference after injection. Right: re-entry and final task success.

commands. The backbone is π_0 -base (PaLiGemma 2B encoder, Gemma 300M action expert, flow matching) [17]; each task is fine-tuned for 5k steps from the public checkpoint on 50 VR-teleoperated demonstrations (Fig. 2).

Four tasks cover complementary primitives (Fig. 8): pick-and-place (prehensile), block pushing (non-prehensile), drawer opening (constrained contact), and whiteboard wiping (sustained contact).

Each task is tested under nominal execution and two OOD families ($N=20$ trials per cell): a random-direction EEf shift (robot-state) and an object displacement (world-state), both at a task-specific phase and magnitude. Directions are randomized per trial but paired between π_0 and π_0 +DUET.

b) Results.: Table VII shows that the transfer preserves nominal success (both 90.0%). Under robot-state EEf shifts, matching the simulation teleport protocol, π_0 falls to 10.0% while DUET recovers to 77.5% (+67.5 pp; +65–70 pp on

TABLE V

FUSION ABLATIONS, SUCCESS (%). FOUR-SUITE ROWS USE $N=2000/2000/4000$; STATIC GATES USE MATCHED OBJECT PROTOCOLS ($N=500$).

Controller	Gate	Clean	Robot displ.	Action exec.
<i>Four-suite average</i>				
DUET	adaptive α_t	96.9	61.7	64.6
DUET-Rel	$\alpha_t=0$	93.8	40.0	27.9
<i>Object suite, matched protocol</i>				
DUET	adaptive α_t	99.2	76.0	84.4
DUET-Rel	$\alpha_t=0$	99.2	36.2	48.8
Absolute-only	$\alpha_t=1$	5.2	1.1	0.8
Fixed fusion	$\alpha_t=0.5$	46.2	39.4	41.0

TABLE VI

COMPONENT ABLATIONS, SUCCESS (%). CFG USES OBJECT ($N=500$); RETRAINS USE LIBERO-10 ($N=500/500/1000$).

Variant	Clean	Robot displ.	Action exec.
<i>CFG schedule, Object suite</i>			
Full DUET	99.2	76.0	84.4
No adaptive CFG	88.2	52.6	57.2
<i>Training replacements, LIBERO-10</i>			
Full DUET	91.8	60.8	63.8
RVQ $\rightarrow$ MLP	0.0	0.0	0.0
Symmetric augmentation	86.8	58.6	60.9

every task). World-state object shifts remain harder: 6.3% vs. 58.8% (+52.5 pp). Figure 9 shows the same contrast on block pushing: π_0 continues off-manifold or pushes empty space; DUET re-enters and completes. The ordering matches the mechanism bound: D_t is strongest when the live pose leaves $\mathcal{M}_l$ and weaker on object-scene change (Fig. 5). The four-task nominal mean is unchanged (block pushing -5.0 pp, wiping $+5.0$ pp).

G. Discussion

The results bound rather than merely confirm the mechanism. D_t ranks failures and drives recovery when the live pose leaves $\mathcal{M}_l$; it is weaker on world-state change and uninformative under invalid images, which the validity guard handles. On arm-state OOD, the gate—not co-training—produces the gain (StarVLA $\approx$ DUET-Rel in simulation; $\pi_0 \ll \pi_0 + \text{DUET}$ on the Franka). Under visual dropout the two sources separate: asymmetric training lifts Rel above StarVLA, and the guard adds further improvement on longer outages. The τ - k plateau is broad enough to avoid per-suite tuning, and the same operating point transfers to π_0 without retuning.

V. LIMITATIONS

DUET assumes absolute state labels aligned with relative action chunks—natural in LIBERO and many robot logs, but may require calibration on other hardware. The gate uses min-max state statistics as a coarse S_t proxy, which may be insufficient in highly multi-modal workspaces. The stateless absolute branch supports recovery but may lose accuracy

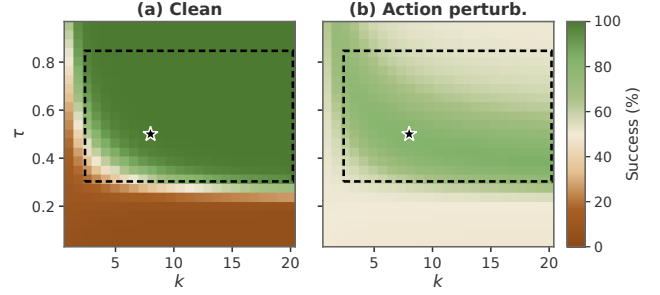


Fig. 7. Object sensitivity to τ, k from 500 clean and 500 perturbed traces. Rectangle: clean $> 90\%$, OOD $> 75\%$; star: operating point.

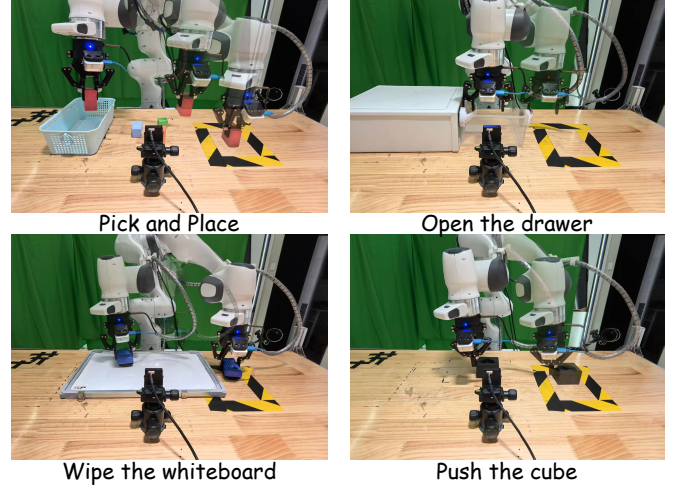


Fig. 8. Real-robot tasks on a Franka Emika Panda (top-left to bottom-right): drawer opening, whiteboard wiping, block pushing, and pick-and-place.

when the current robot state carries task information not visible in images. Two scope limits follow: D_t is weaker on world-state change (Fig. 5(b); Table VII), and both heads degrade together under a fully invalid observation, leaving only the cached-waypoint guard. Irreversible failures and tasks requiring a new strategy are out of scope.

VI. CONCLUSION

We presented DUET, a dual-branch VLA policy for endogenous failure detection and recovery. By aligning a local relative-action tangent field with a stateless, RVQ-anchored absolute-waypoint manifold, DUET uses cross-space disagreement to detect when execution becomes OOD and to activate a closed-loop recovery direction. This unifies failure detection and recovery without an external detector or additional recovery demonstrations. Because the module attaches only to the shared vision-language representation, the identical design transfers from StarVLA in simulation to π_0 on a real Franka arm without backbone-specific adaptation.

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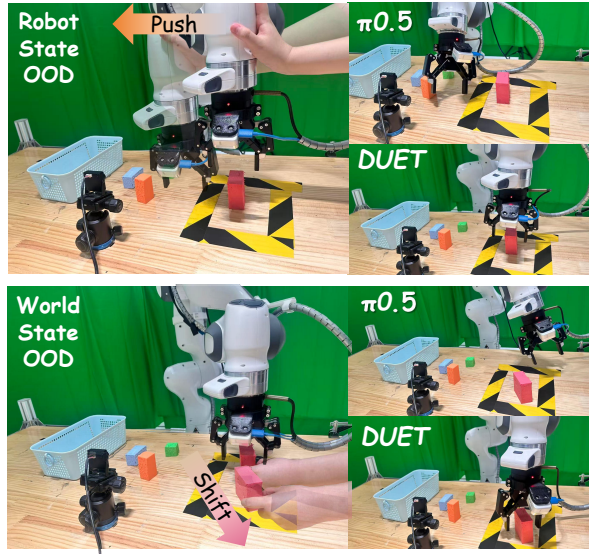


Fig. 9. Qualitative recovery on block pushing. Top: robot-state EEF shift— π_0 fails (0), DUET recovers. Bottom: world-state cube shift— π_0 pushes empty space (0), DUET reacquires the object.

TABLE VII
REAL-ROBOT SUCCESS (%) WITH A π_0 -BASE BACKBONE ($N=20$
TRIALS PER CELL; OVERALL POOLS $N=80$). ROBOT-STATE:
RANDOM-DIRECTION EEF SHIFT. WORLD-STATE: OBJECT
DISPLACEMENT. Δ : π_0 +DUET MINUS π_0 .

Task	Nominal		Robot-state (EEF shift)			World-state (Obj. shift)		
	π_0	+DUET	π_0	+DUET	Δ	π_0	+DUET	Δ
Pick-and-place	95.0	95.0	15.0	85.0	+70.0	5.0	50.0	+45.0
Block pushing	95.0	90.0	10.0	75.0	+65.0	0.0	60.0	+60.0
Drawer opening	95.0	95.0	10.0	80.0	+70.0	20.0	70.0	+50.0
Whiteboard wiping	75.0	80.0	5.0	70.0	+65.0	0.0	55.0	+55.0
Overall	90.0	90.0	10.0	77.5	+67.5	6.3	58.8	+52.5

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